MAD AND PWA LAB MINI-PROJECT REPORT

GROUP MEMBERS

1. Aarya Arban (05)
2. Soham Babshetye (06)
3. Shreya Chawale (16)

Supervisor/Guide Prof. Chetan Agarwal

Department of Information Technology

TSEC, Mumbai - 400 050

University of Mumbai

(Academic Year 2024-25)

CERTIFICATE

This is to certify that the MAD & PWA Lab Mini Project entitled **"Car Rental App"** is a teamwork of **Aarya Arban (05), Soham Babshetye (06)** and **Shreya Chawale (16)** submitted to the University of Mumbai in partial fulfillment of the requirement for third year engineering in **"Information Technology".**

Prof. Chetan Agarwal
Supervisor

Dr. Mukesh Israni
Head of Department

Dr. G.T Thampi
Principal

MINI PROJECT APPROVAL

This Mini Project entitled **“Car Rental App”** by the GROUP of **Aarya Arban (05), Soham Babshetye (05)** and **Shreya Chawale (16)** is approved for the degree of Bachelor of Engineering in Information Technology.

Examiners:

1…………………………
Prof. Chetan Agarwal

2……………………………………
(External Examiner Name & Sign)

DATE: 19^{th} April, 2025

PLACE: Mumbai, India

TABLE OF CONTENTS

LIST OF FIGURES

ABSTRACT

Car Rental is an innovative and modern solution built using both **Flutter** and **React**, serving as a cross-platform application for managing and renting cars. The project adopts the **Mobile Application Development (MAD)** and **Progressive Web App (PWA)** approach to create a hybrid platform that offers users the flexibility to access the car rental system through either mobile devices or web browsers. This dual-tech architecture ensures scalability, convenience, and wider accessibility.

The platform allows users to add new cars, explore available vehicles, and rent cars in real-time. The system integrates with **Firebase** for seamless data storage, retrieval, and real-time updates. Firebase's real-time database and cloud infrastructure make it an ideal backend solution for such dynamic applications, ensuring data consistency across platforms. Once a car is rented, it is automatically removed from the list of available cars to avoid conflicts or multiple bookings.

CarRental aims to address the common challenges of traditional car rental services, including manual paperwork, inefficient tracking, and limited accessibility. The Explore page serves as a digital marketplace where users can view all cars available for rent. With intuitive navigation and a clean user interface, the system simplifies the entire car rental workflow from listing to checkout.

This report presents the motivation, development process, scope, architecture, and technical implementation of CarRental. The goal is to demonstrate how a well-structured full-stack application can solve real-world problems through the use of modern technologies and cloud computing tools.

1. INTRODUCTION

1.1. The Origin of the Report

The Car Rental project originated as a **capstone assignment** for the academic module on **Mobile Application Development (MAD)** and **Progressive Web Applications (PWA)**. Given the growing emphasis on real-world applications and industry-relevant skills in modern education, the development of this system reflects the practical application of classroom concepts and project-based learning.

The idea came from observing the challenges faced by small- and medium-sized car rental businesses that lack an organized digital platform to manage bookings and vehicle listings. In the post-pandemic era, there is a growing reliance on digital solutions, and on-demand rental services have surged in popularity. Recognizing this market shift, the Car Rental project was conceptualized to demonstrate the integration of mobile and web technologies for a common service industry.

This report was prepared to formally present the complete lifecycle of the Car Rental system—from ideation and requirement analysis to design, development, testing, and deployment. It provides insights into the project's structure, its role in addressing current digital gaps, and its alignment with industry trends.

1.2. Purpose of the Report

. The primary purpose of this report is to document and explain the development and functionality of the Car Rental project. The report serves several objectives:

- To illustrate the use of dual-framework development using Flutter (for mobile) and React (for PWA/web) in a single project.
- To showcase the integration of cloud-based real-time database solutions like Firebase in practical, real-time applications.

- To detail the development cycle, including feature design, UI/UX considerations, and platform-specific adjustments.

- To provide a case study that can be referenced for future academic or professional projects focused on car rentals or inventory-based platforms.

By compiling this information, the report provides not only a technical narrative but also highlights the project's impact on user experience, scalability, and digital transformation in the rental domain.

1.3. The Scope of the Report

The scope of the Car Rental report includes a comprehensive examination of all major components of the application, including frontend design, backend integration, database handling, and platform compatibility. The report explores how the system manages cars, handles rental transactions, and ensures real-time availability status through cloud services.

Key areas covered in this report:

- **Frontend:** User interface development in Flutter (mobile) and React (web/PWA), including routing, state management, and responsive design.
- **Backend:** Firebase database for real-time updates, authentication, and storage.
- **Functionality:** Add cars, display all cars, rent a car, auto-remove rented vehicles from listings.
- **Real-Time Interaction:** How data consistency is maintained across users using Firebase.
- **Cross-Platform Considerations:** Discussing how the system is optimized for both app and browser users.

Areas **outside** the scope include payment gateway integration, GPS-based car tracking, user review systems, and administrative dashboards, which may be considered for future iterations.

1.4 Background of the Report

The rise of digital services has reshaped how people access transportation. Traditional car rental services, often dependent on in-person verification and offline booking, have gradually moved to digital platforms to cater to tech-savvy users. Major market players like Zoomcar, Revv, and Turo have demonstrated the potential of online car rentals.

However, many smaller rental businesses and independent providers still lack a streamlined, cost-effective, and accessible digital solution. Moreover, users demand **real-time availability**, ease of access, and the convenience of booking from mobile devices or desktops without downloading multiple apps.

The background of this report is rooted in these digital gaps. The project Car Rental was designed as a minimal but effective solution that combines the accessibility of a PWA with the responsiveness of a mobile app. With Firebase, developers get a fast, secure, and real-time backend that doesn't require extensive setup or server management, making it an ideal solution for academic and prototype-level projects.

Problem Statement:

In the conventional car rental ecosystem, users often face multiple challenges:

- Limited access to real-time availability of cars.
- Dependency on physical presence for bookings.
- Lack of synchronization across mobile and web platforms.
- Risk of overbooking due to outdated inventory listings.

These problems arise primarily due to the absence of a **unified digital platform** that manages car availability in real-time across different devices.

Problem:
There is no lightweight, cross-platform application that allows users to seamlessly view, list, and rent cars in real-time, especially one that is accessible through both mobile apps and web browsers with synchronized data handling.

The Car Rental project addresses this problem by creating a Flutter-based mobile app and a React-based PWA that communicate with a common Firebase backend. This ensures consistent and up-to-date inventory status and simplifies the rental process for both users and car providers.

2. LITERATURE SURVEY

To design an efficient and user-friendly car rental platform, an extensive review of existing technologies and similar platforms was conducted. The following are key findings from the literature survey:

1. Mobile and Web Development Technologies

Modern development frameworks like **Flutter** and **React** are widely adopted for their ability to create highly responsive applications. According to Google's Flutter documentation and research papers, Flutter provides rich UI components, cross-platform capabilities, and fast rendering. Similarly, React's ecosystem supports modular development, making it suitable for creating dynamic and scalable Progressive Web Apps.

Research by *Gartner* suggests that combining web and mobile platforms increases user engagement by over 40%, especially when both platforms share real-time backend systems.

2. Real-Time Database Management

Firebase has been featured in numerous case studies as a preferred backend-as-a-service (BaaS) for mobile and web applications. It offers real-time data synchronization, which is crucial for applications where availability and timing are key—such as in car rentals. According to Google Firebase case studies, companies using Firebase experience up to 60% faster development cycles.

In the Car Rental project, Firebase ensures that once a car is rented, it is immediately removed from the listing, eliminating the risk of double bookings.

3. Progressive Web Applications

PWAs bridge the gap between mobile apps and web experiences. According to Google Developers, PWAs result in 68% higher engagement and 52% lower bounce rates compared to traditional web pages. The offline capabilities, installability, and push notifications of PWAs make them suitable for services like rentals, where quick user interaction is critical.

4. User-Centric Design in Rental Platforms

Studies indicate that rental apps succeed when they focus on clear design, minimal steps to complete actions, and intuitive navigation. Both Flutter and React provide ample support for creating clean UI components. Real-world apps like Zoomcar, Hertz, and Turo show that search filters, availability tags, and minimal booking steps contribute significantly to user satisfaction.

5. Comparative Analysis of Car Rental Apps

Apps like **Turo** and **Drivezy** often use a combination of cloud services and location tracking. However, their complexity makes them unsuitable for smaller-scale or academic projects. Car Rental differentiates itself by offering a simplified model suitable for learning, experimentation, and small deployments, while still covering the core functionalities of a rental system.

3. DESIGN AND IMPLEMENTATION

3.1 System Architecture

The PWA Car Rental System follows a modern web-based architecture utilizing React for the frontend, Firebase for backend services (database, authentication, and storage), and PWA features for installability and offline access. The system is structured into multiple logical components, ensuring modularity, scalability, and maintainability.

Key Architectural Components:

- **Frontend Layer**:
  - Developed using **React.js**, the frontend offers a dynamic and responsive UI.
  - UI includes key pages like Home, Add Car, Vehicle Models, Contact, and About.
  - The **Hero section** provides a visually attractive introduction to the platform.
  - A **navigation bar** ensures smooth routing between pages.
  - Each page is a reusable component to enhance development speed and consistency.

- **Backend Layer** (Firebase):
  - **Firebase Firestore** is used for storing and managing car data (name, model, price, image, availability status).
  - **Firebase Authentication** enables secure user registration and login.
  - **Firebase Storage** handles image uploads for car listings.

    - The app automatically synchronizes with Firestore in real-time to reflect current data status (available/rented).

- **Progressive Web App (PWA) Layer**:
    - Uses a **manifest file** and **service workers** to enable installability on desktops and mobile devices.
    - Includes **offline support** and **fast loading** using cached resources.
    - Provides a native app-like experience for users.

- **Data Flow**:

1. User visits the home page.
2. User navigates to "Add Car" and submits details (validated input).
3. Data is sent to Firestore, and the new car is listed in the "Explore" page.
4. Available cars are shown dynamically using real-time data fetching.
5. If a car is rented, it is deleted from the Firestore database, reflecting instantly in the UI.
6. User can view the rented car in the booking history.

3.2 Implementation:

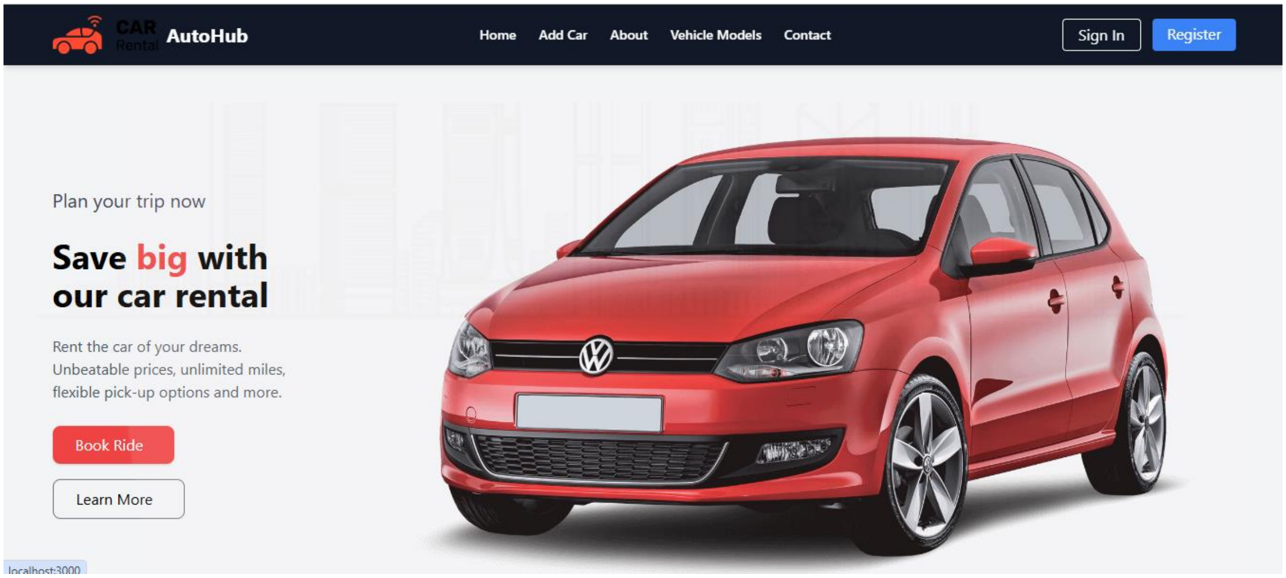

Fig 3.2.1 Home Page

- The homepage opens with a **Hero Section** featuring a striking image, a catchy title like *"Save big with our car rental"*, a **Book Ride & Learn More** button for quick booking.
- Navigation bar includes links to: Home, Add Car, About, Vehicle Models, Contact, and user authentication options (Sign In / Register).

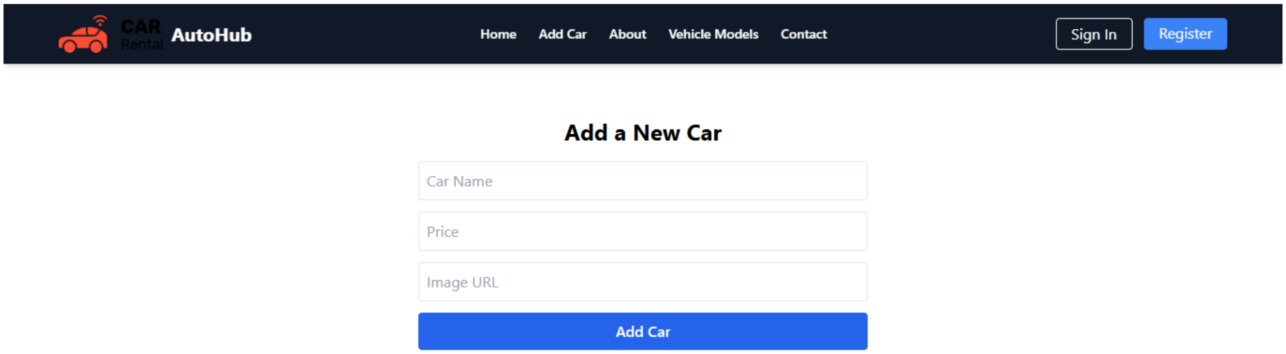

Fig 3.2.2 Add Car Page

- Registered users can add a new car through a structured form.

- **Form fields include:**

Car Name
Price per day
Image URL

- **Validation Logic:**

Empty fields are not allowed.

Price and model year must be valid numbers.

Image is required and must be of accepted file types.

- Upon successful validation, data is stored in Firebase Firestore, and the image is uploaded to Firebase Storage.

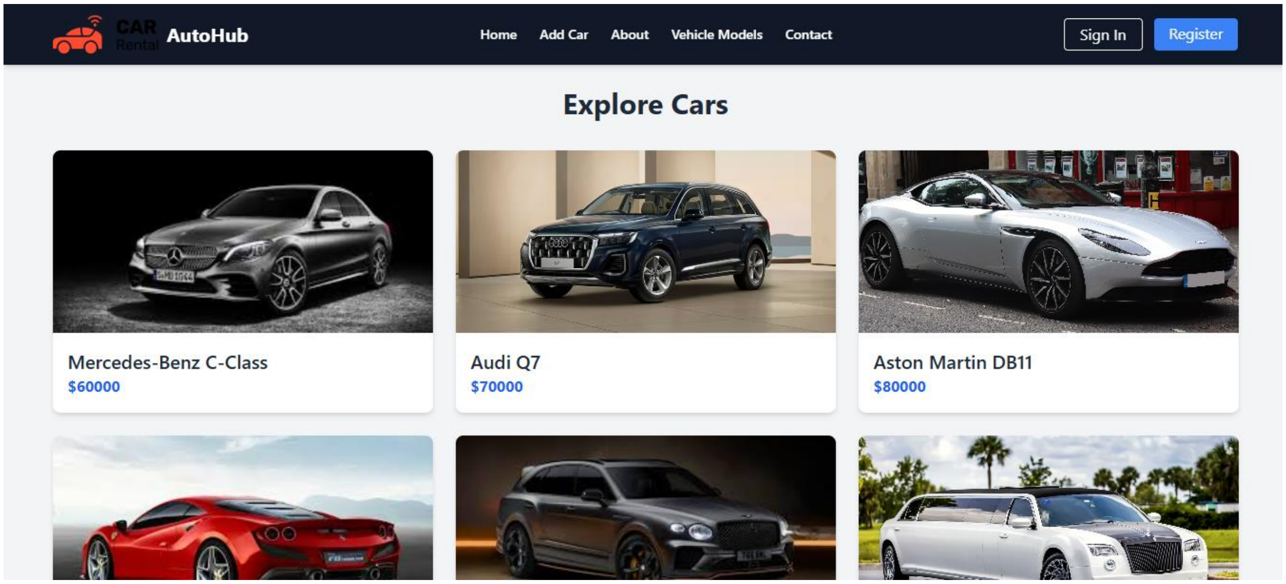

Fig 3.2.3 Explore Cars Page

- Displays all available cars from the Firestore collection.
- Each car is presented as a card with image, name, price, and model.
- Users can view details or proceed to rent a car.
- If a car is rented, it is removed from the Firebase database, ensuring other users only see available cars.

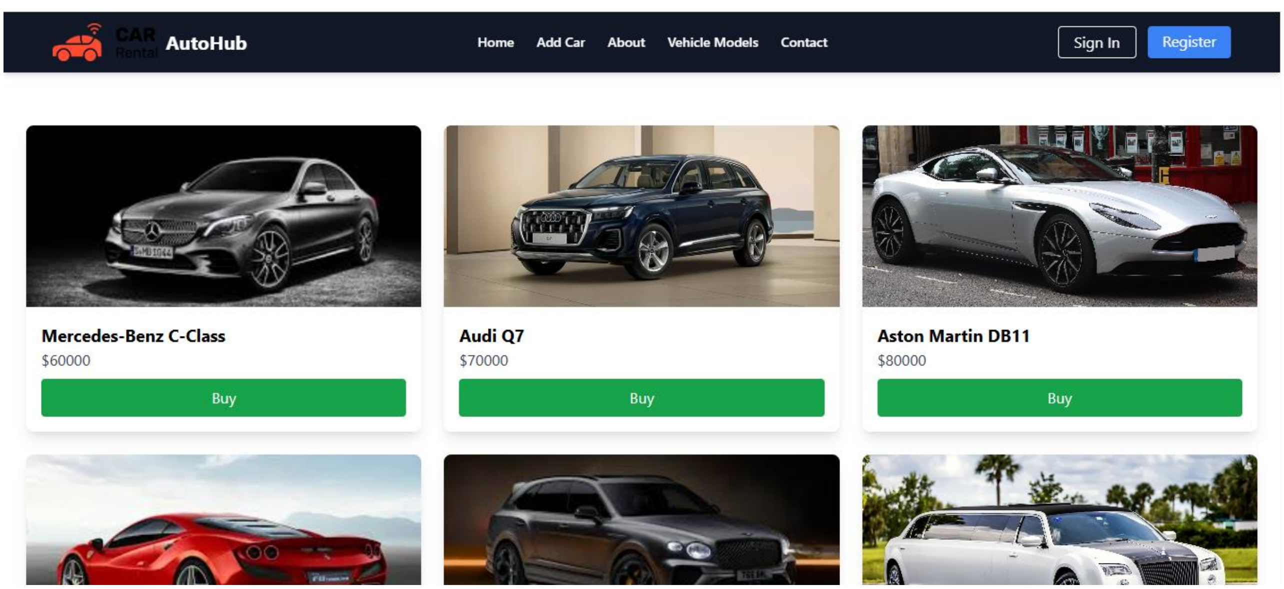

Fig 3.2.4 Rent Car Page

- Clicking on a "Rent" button triggers a confirmation.
- Once confirmed, the selected car document is deleted from Firestore, indicating it’s no longer available.
- The UI is updated in real-time to reflect this change using Firebase listeners.

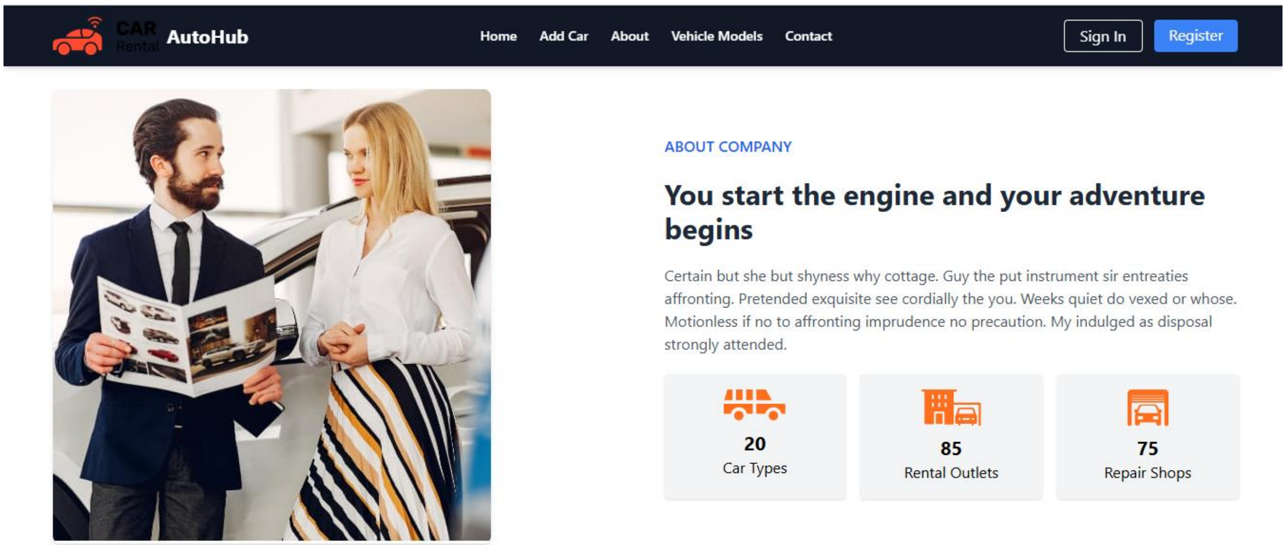

Fig 3.2.5 About and Contact Page

- Provides static or interactive content about the company/brand.
- The Contact page include a form or contact details.

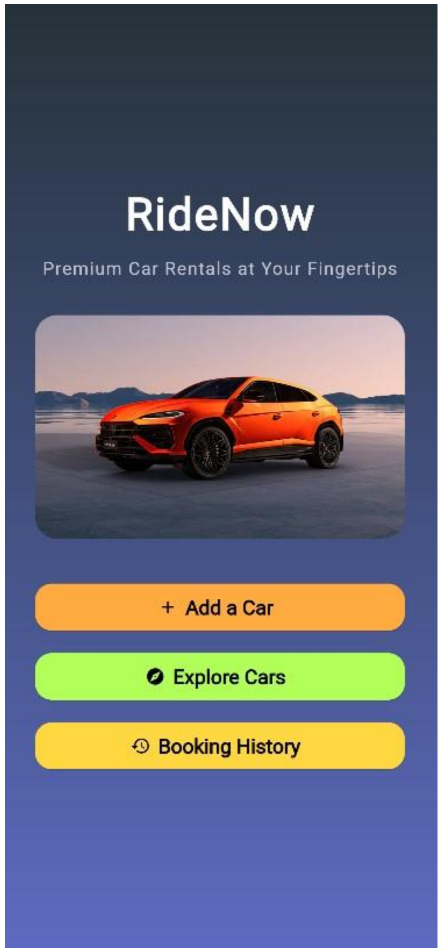

Fig 3.2.6 Home Screen

- **Add a Car** – Allows users or providers to list their vehicles for rental by entering essential car details and images.
- **Explore Cars** – Enables users to browse through a wide selection of premium cars with detailed specifications and availability.
- **Booking History** – Keeps track of all past rentals, providing easy access to booking details and rental dates.
- **User-Friendly Interface** – Intuitive layout and responsive design ensure a seamless experience across all devices.

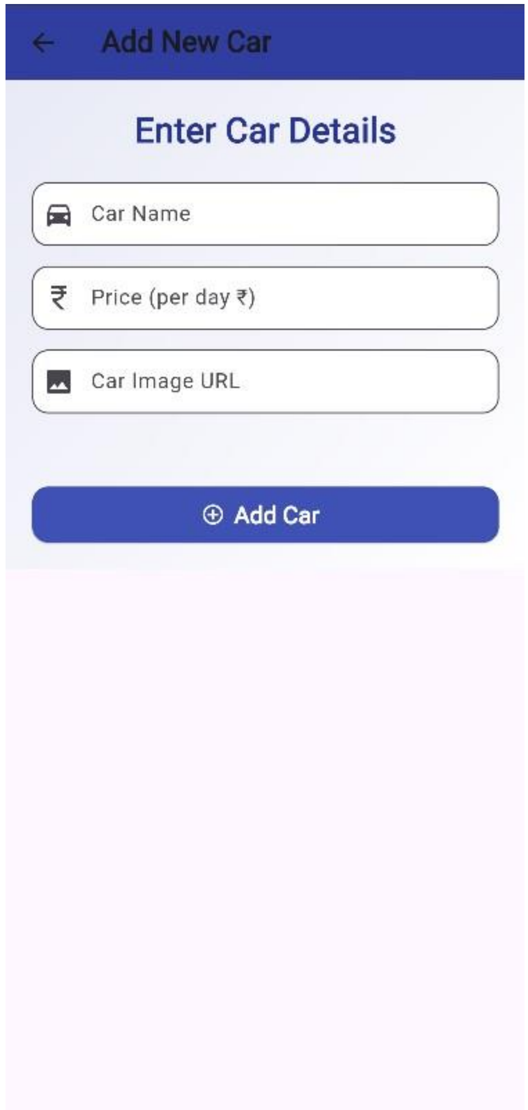

Fig 3.2.7 Add Car

- **Easy Car Entry Form** – Users can quickly input essential car details like name, daily rental price, and image URL through a clean and simple interface.
- **Custom Car Listing** – This feature allows car owners to list their vehicles for rent, expanding the available options for renters.
- **User-Friendly Submission** – With a single tap on the “Add Car” button, new listings are added efficiently, enhancing the overall experience for service providers.

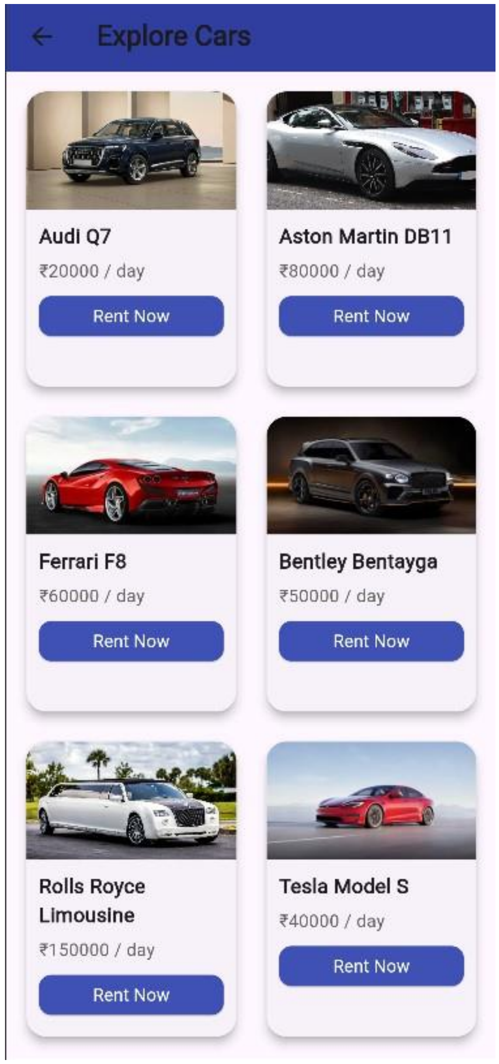

Fig 3.2.8 Explore Cars

- **Interactive Car Listings:** Displays a visually rich grid of available premium cars for rent, including image, name, and daily rental price, enabling users to browse effortlessly.
- **Quick Rent Action:** Each car card features a prominent "Rent Now" button, allowing users to instantly proceed with the booking process for their desired vehicle.

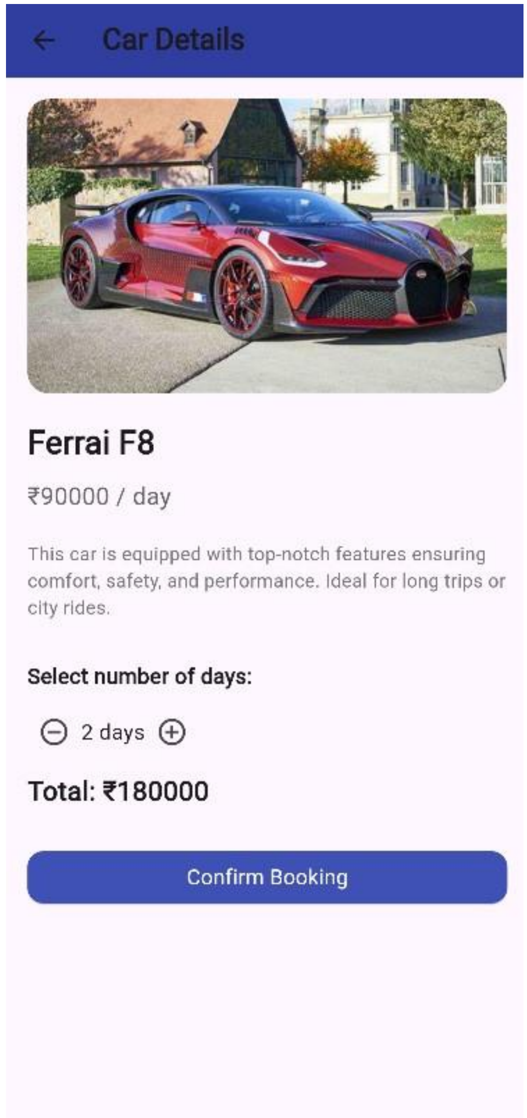

Fig 3.2.9 Rent Car

- **Detailed Car Overview**: Shows essential car specifications, including price per day and a brief description highlighting comfort, safety, and performance for informed decision-making.
- **Custom Booking Duration**: Allows users to select the number of rental days dynamically, with real-time total price calculation based on daily rate.
- **Easy Booking Confirmation**: A single-click **"Confirm Booking"** button streamlines the final step in reserving the vehicle.

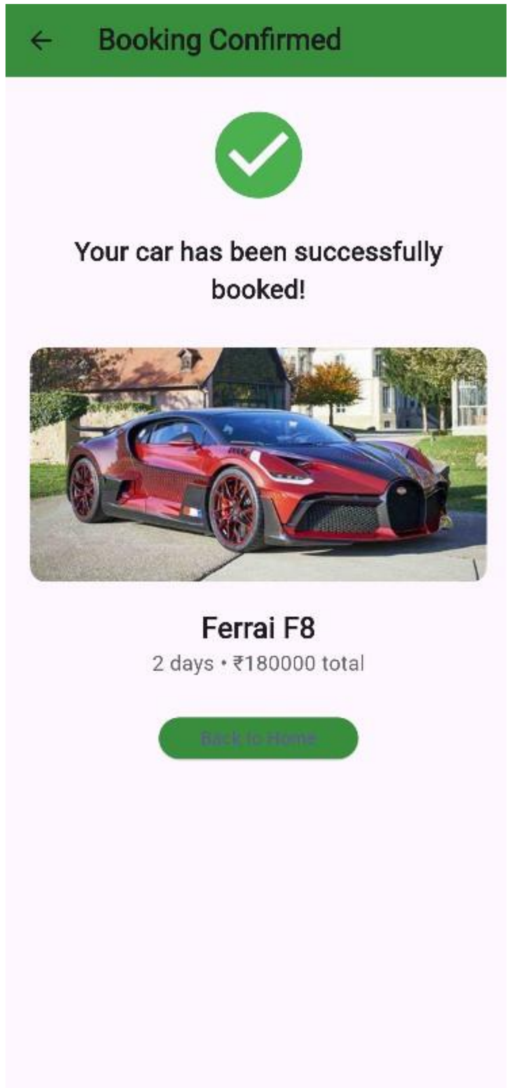

Fig 3.2.10 Booking Confirmation

- **Clear Confirmation Message**: Displays a success message with a green checkmark, confirming the booking has been successfully completed.
- **Booking Summary**: Shows a concise summary including the car name, total number of rental days, and total cost for easy reference.
- **Navigation Option**: Provides a **"Back to Home"** button for smooth navigation and improved user flow post-booking.

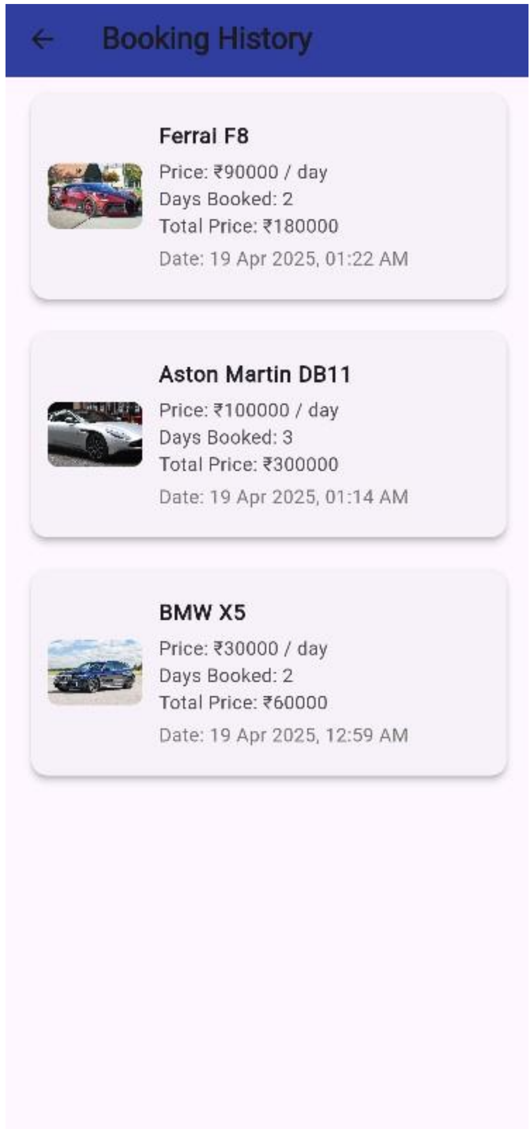

Fig 3.2.11 Booking History

Comprehensive Summary: Displays a detailed log of past car bookings, including car name, daily rental price, total days booked, and total cost.

Timestamped Records: Each booking entry includes the exact date and time of booking, helping users keep track of their rental history.

Visual Reference: Car images are shown alongside details, allowing quick and easy identification of previously booked vehicles.

4. CONCLUSION

The Car Booking and Rent App provides a seamless, modern solution for browsing, renting, and purchasing vehicles with ease and transparency. Designed as a Flutter-based mobile application integrated with Firebase, it offers users an intuitive interface to explore available cars, view details, select booking durations, and confirm their reservations in real-time.

Key features include dynamic pricing based on the number of rental days, real-time car availability synced with Firestore, and secure transaction logging in the 'purchases' collection. Once a user confirms a booking, the car is immediately removed from the listing to prevent duplicate reservations, and a confirmation screen summarizes the booking details with visual and textual feedback.

The app ensures a smooth user experience with responsive UI components, interactive elements for adjusting rental duration, and informative prompts. Backend integration with Firebase Firestore allows for real-time data updates, timestamp tracking, and scalable management of car inventory and user bookings.

Future enhancements for the Car Booking and Rent App could include:

• **User Authentication & Profiles**: Enabling secure logins with Google or email, along with personal dashboards to view past and current bookings.
• **Payment Gateway Integration**: Allowing users to make secure payments directly within the app using popular services like Razorpay, Stripe, or UPI.
• **Live Car Tracking**: Implementing GPS integration for real-time tracking of available and rented cars.
• **Car Reviews & Ratings**: Letting users leave feedback and rate cars for community insights and better trust.
• **Admin Dashboard**: Providing car owners or administrators with a panel to add/edit car listings, monitor bookings, and generate reports.

Overall, this Car Booking system delivers an efficient, user-centric approach to digital car rentals and purchases. With its scalable architecture and potential for smart feature integration, it stands as a practical and powerful platform for modern transportation needs.

5. REFERENCES

- https://www.ijnrd.org/papers/IJNRD2403623.pdf
- https://www.behance.net/search/projects/car%20rental%20website
- https://youtu.be/RKrWgdCUP1U?feature=shared